

Exploring Used Car Prices

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1. Introduction

For this project, I used the Used Cars dataset, which contains thousands of car sales across the United States. After cleaning, the dataset includes 128,752 records. Each record features details like price, year, manufacturer, model, fuel type, and mileage (odometer). My goal was to answer three questions: (1) What are the main factors that determine a car's price? (2) How does mileage affect the value of a car? (3) Which manufacturer groups have the highest resale prices?

2. Cleaning Summary

The raw dataset had many problems that needed fixing before I could analyze it:

- **Missing values & Duplicates:** I selected the 8 most important columns: price, year, manufacturer, model, fuel, transmission, odometer, and condition. I removed all rows with missing values and deleted duplicate rows to ensure the data was clean.
- **Data types:** I changed the year column to an integer and the odometer column to a decimal (float) to make them better for math calculations.
- **Outliers:** I found many unrealistic prices, I removed all cars priced below \$500 and capped the price at the 99th percentile to remove extreme values. I also removed cars with 0 mileage because it is illogical for used cars.

I wrapped all these steps into a `clean_data()` function and added checks to verify that there are no nulls and the column count is correct (8 columns).

3. Feature Engineering Summary

I created several new features to help the analysis:

- **car_age = 2026 - year:** This is more useful than the year because price depends on how old the car is.
- **mileage_per_year = odometer / (car_age + 1):** This shows how much the car was driven every year. I used "+1" to avoid dividing by zero for new cars.
- **condition_score:** I converted the condition words into numbers from 0 to 5. This helps the computer understand that "new" is better than "fair".
- **One-hot encoding:** I changed Fuel and Transmission categories into numbers (0 and 1).
- **Log-transformed Price:** The price was very skewed. I applied `np.log1p()` to make the distribution better for analysis.

- **Scaling:** I used StandardScaler on odometer and car_age to put them on the same scale.

4. Key Findings

- **Finding 1 — Car Age is the strongest factor.** The correlation map showed that as car age increases, the price drops quickly. Most cars lose a lot of value in the first few years.
- **Finding 2 — Mileage vs Price.** The scatter plot showed that cars with lower mileage have higher prices, but once a car passes 150,000 miles, the price becomes very low regardless of the age.
- **Finding 3 — Group Gaps.** The analysis revealed that modern cars (after 2010) are nearly three times more expensive on average than old cars (before 2000), which makes sense.

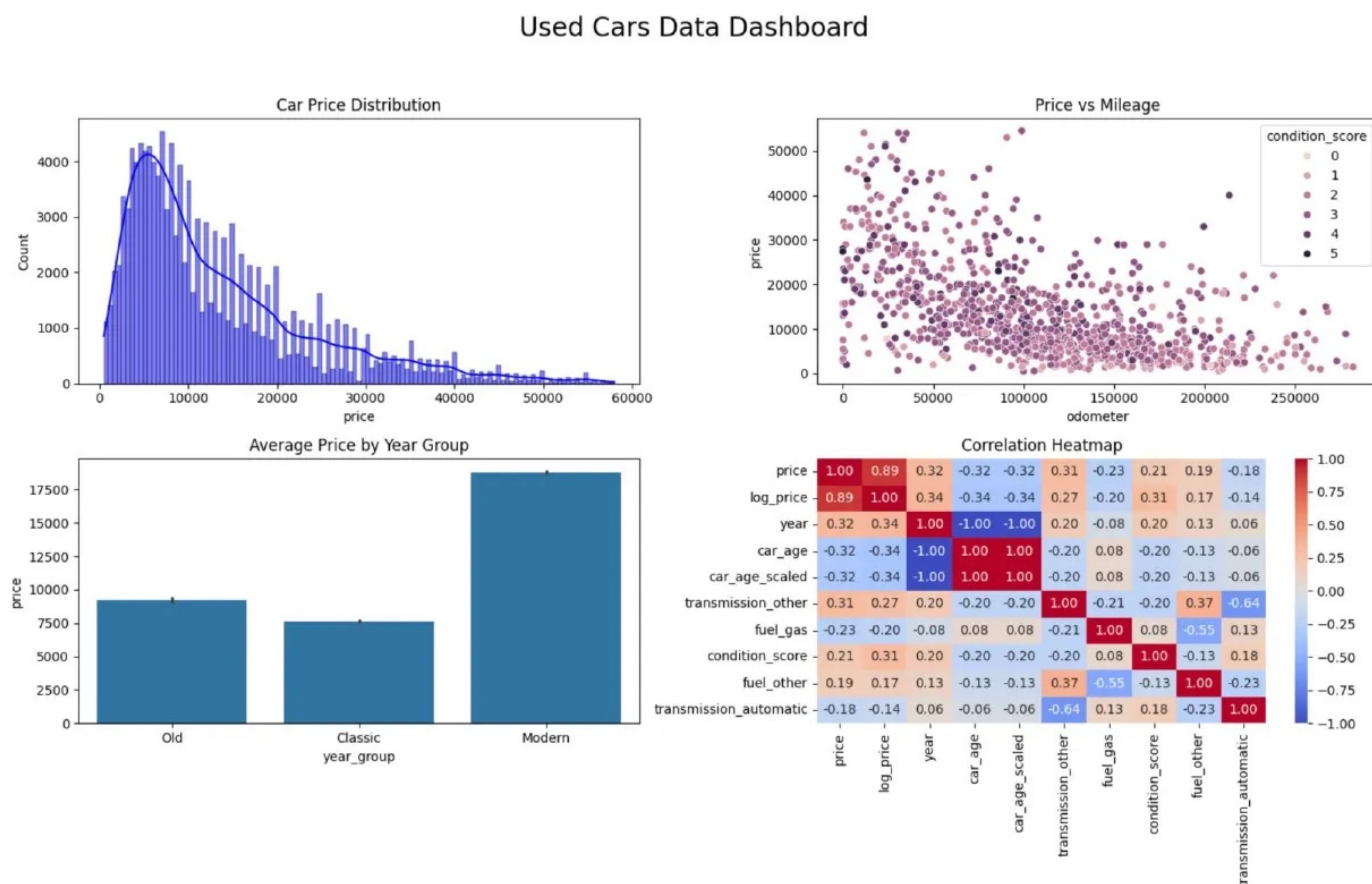


Figure 1: Used Cars Data Dashboard

5. What I Would Do Next

If I had more time, I would: (1) create a model to predict the car price based on the features I created; (2) check if the car's Location (country - city) changes the price; and (3) look at which car colors or models sell faster than others.